

8.3.16

AI25BTECH11036-SNEHAMRUDULA

Question:- An arch is in the form of a semi-ellipse. It is 8 m wide and 2 m high at the centre. Find the height of the arch at a point 1.5 m from one end?

Solution:

We are given a semi-ellipse of width 8 m and height 2 m at the centre.

Step 1: Standard Form

For an ellipse, the equation is

$$\mathbf{q}^T \mathbf{V} \mathbf{q} = 1 \quad (0.1)$$

where

$$\mathbf{V} = \begin{pmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{b^2} \end{pmatrix} \quad (0.2)$$

Here, the semi-major axis $a = 4$ (half of 8 m) and semi-minor axis $b = 2$.

Thus,

$$\mathbf{V} = \begin{pmatrix} \frac{1}{16} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \quad (0.3)$$

Step 2: Equation of Ellipse

Let

$$\mathbf{q} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (0.4)$$

Substituting into (1),

$$\frac{\alpha^2}{16} + \frac{\beta^2}{4} = 1 \quad (0.5)$$

Step 3: Position at 1.5 m from End

Since the width is 8 m, the end corresponds to $\alpha = \pm 4$. At a point 1.5 m from the end, the coordinate is

$$\alpha = 4 - 1.5 = 2.5 \quad (0.6)$$

Step 4: Solve for Height

Substitute $\alpha = 2.5$ in (6):

$$\frac{(2.5)^2}{16} + \frac{\beta^2}{4} = 1 \quad (0.7)$$

$$\frac{6.25}{16} + \frac{\beta^2}{4} = 1 \quad (0.8)$$

$$\frac{\beta^2}{4} = 1 - 0.390625 \quad (0.9)$$

$$\frac{\beta^2}{4} = 0.609375 \quad (0.10)$$

$$\beta^2 = 2.4375 \quad (0.11)$$

$$\beta \approx 1.561 \quad (0.12)$$

Final Answer

Hence, the height of the arch at a point 1.5 m from the end is approximately

$$\beta \approx 1.56 \text{ m} \quad (0.13)$$

